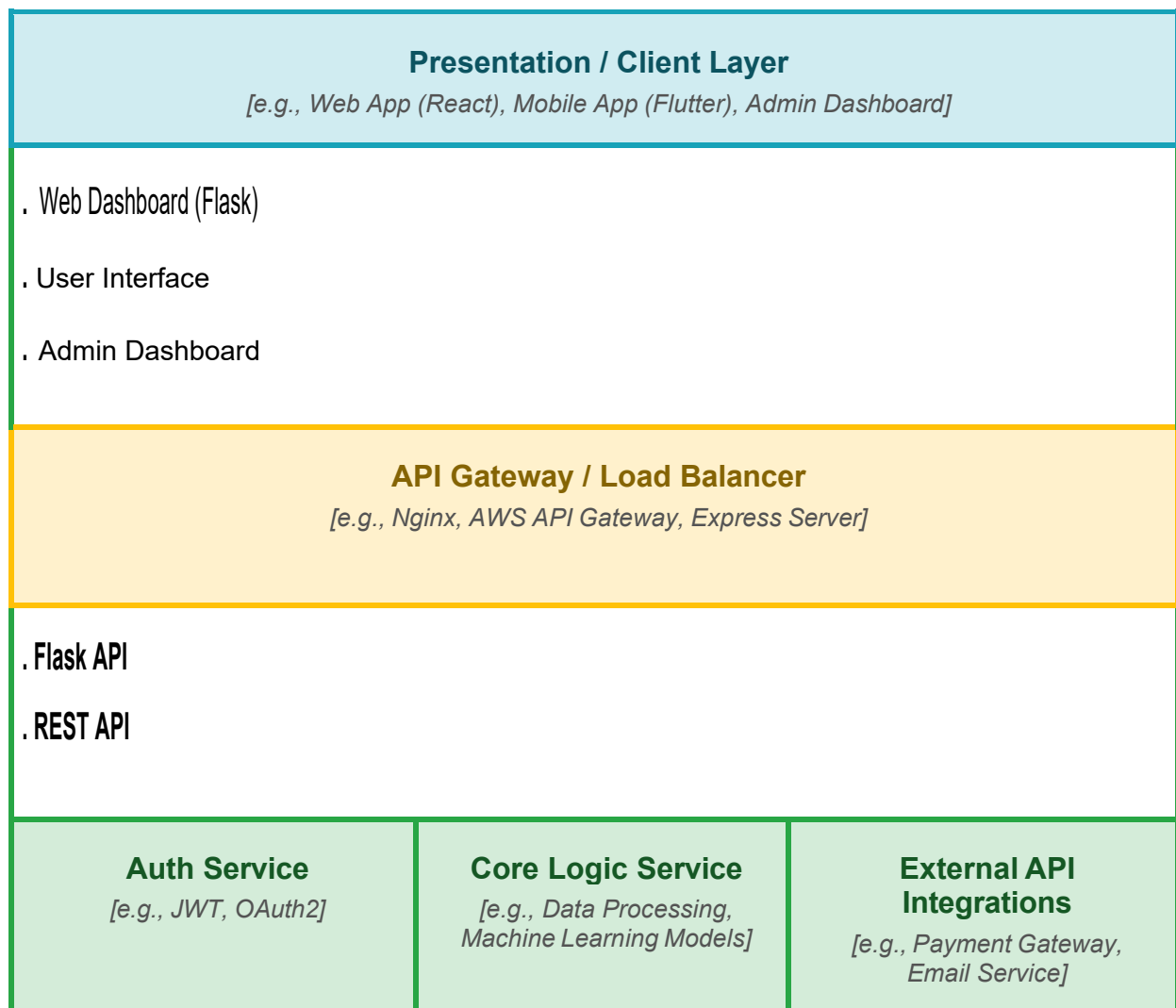


## Solution Architecture

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 10 July 2026                            |
| <b>Team ID</b>       | XXXXXX                                  |
| <b>Project Name</b>  | Rising Waters – Flood Prediction System |
| <b>Maximum Marks</b> | 5 Marks                                 |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



|                        |                       |                                |
|------------------------|-----------------------|--------------------------------|
| . User Login           | . Data Preprocessing  | . weather Api (Optional)       |
| . Admin Authentication | . feature Engineering | . Rainfall Data API (Optional) |
| Machine Learning Model |                       |                                |

### **Data / Storage Layer**

*[e.g., PostgreSQL Database, MongoDB, AWS S3]*

. Flood Prediction dataset(Csv)

. SQLite / Database Mysql

## Component Description Table

| Component Name              | Description / Role in Architecture                                                              | Technologies Used                    |
|-----------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------|
| <i>[Presentation Layer]</i> | Provides a web interface for users and admins to view flood predictions and reports.            | Flask,HTML,CSS                       |
| <i>[API Gateway]</i>        | Handles requests between the user interface and the machine learning model.<br>Flask REST API   | Flask REST API                       |
| <i>[Microservices]</i>      | Performs data preprocessing, model training, and flood prediction using machine learning.       | Python, Scikit-Learn, pandas , numpy |
| <i>[Database]</i>           | Stores flood dataset, prediction results, and trained model information.<br>CSV Dataset, SQLite | Csv Dataset,sqLite                   |